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Multi-Objective Reinforcement Learning Using Routed Ensembles and Trajectory Attribution

Objective-Conflict Coefficients Quantify Preference Trade-offs, While Routed Ensembles and Kernel-Surrogate Attribution Close the Multi-Objective Efficiency Gap

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Multi-objective reinforcement learning demands a single agent to simultaneously satisfy multiple reward objectives weighted by a user-supplied preference vector—a task made hard when objectives conflict and gradient directions oppose each other. This paper introduces objective-conflict coefficients that formally quantify this conflict, derives matching online regret bounds, and proposes two practical algorithms: a routed ensemble that routes each trajectory to the basis policy with highest expected return under its preference weight, and a trajectory attribution module based on kernel ridge regression that identifies which trajectories are most valuable for which preference objectives.

AT A GLANCE

Problem: A single preference-conditioned policy fails when objectives strongly conflict; the degree of conflict was not previously formalised.

Why it matters: MORL is central to robotics, resource allocation, and fairness-aware learning, where trade-offs among objectives must be navigated at deployment.

Method: Objective-conflict coefficients + routed ensemble policy learning (trajectory routing to K basis policies) + kernel ridge regression trajectory attribution.

Result: Consistent improvements over single-policy and standard ensemble baselines on multi-objective robotic control benchmarks.

Evidence: Multi-objective MuJoCo locomotion benchmarks; formal regret analysis as a function of maximum conflict coefficient.

The Big Picture

In multi-objective reinforcement learning (MORL), the agent receives a vector reward $\mathbf{r} = (r_1, \ldots, r_K)$ and a preference weight $\mathbf{w} \in \Delta^{K-1}$. The goal is to find a policy π that maximises $\mathbf{w}^\top J(\pi)$ across many possible $\mathbf{w}$, effectively learning the entire Pareto front of preference-weighted returns.

Conflict arises when $\nabla_\theta J_i(\pi_\theta)$ and $\nabla_\theta J_j(\pi_\theta)$ point in opposing directions, forcing any gradient step to damage at least one objective. Prior approaches either concatenate $\mathbf{w}$ to the state (ignoring conflict) or use hand-crafted ensemble partitions (ignoring data efficiency). This work addresses both gaps.

Why Existing Approaches Fall Short

Conditioned single policy: Appending $\mathbf{w}$ to the state does not prevent gradient conflicts. For strongly conflicting objectives, the single policy converges to a suboptimal compromise across all preferences rather than achieving the Pareto front.

Naïve ensembles: Manually partitioning the preference space into K fixed regions assigns trajectories to basis policies by region membership, ignoring that trajectories generated under one preference weight may be highly informative for another.

No formal measure of conflict: Without a quantitative measure of objective conflict, there is no principled way to predict how many basis policies are needed or which trajectory assignments are most beneficial.

Core Mechanism

Routed Ensemble Policy Learning maintains K basis policies $\{\pi_1, \ldots, \pi_K\}$ and iterates:

1. Collect trajectories $\{\tau_j\}$ by sampling basis policies according to the current preference weight distribution.
2. **Route:** Assign each τ_j to the basis policy with the highest expected return under τ_j's preference weight: $k^*(\tau_j) = \arg\max_k \mathbf{w}_j^\top J_k(\pi_k)$.
3. **Update:** Apply PPO to each π_k using only trajectories routed to it.

Trajectory Attribution uses a kernel ridge regression surrogate $\hat{f}(\pi, \mathbf{w}; \mathcal{D})$ to estimate policy performance as a function of the training trajectory set $\mathcal{D}$. Leave-one-out attribution scores ϕ_j^k identify which trajectories help or hurt each basis policy under each preference weight, enabling curriculum sampling.

Technical Formulation

Objective-Conflict Coefficient. For objectives i and j:

$$c_{ij}(\pi_\theta) = \frac{-\nabla_\theta J_i^\top \nabla_\theta J_j}{\|\nabla_\theta J_i\| \, \|\nabla_\theta J_j\|}$$

A positive c_{ij} indicates conflict (gradients oppose); the maximum over all pairs and all preference weights bounds the MORL regret:

$$\text{Regret}(T) = O\Big(\sqrt{T \cdot c_{\max}}\Big)$$

where $c_{\max} = \max_{i \neq j, \mathbf{w}} c_{ij}(\pi^*(\mathbf{w}))$.

Kernel Surrogate. Let $\phi(\tau, \mathbf{w})$ be a feature embedding of trajectory τ and preference weight $\mathbf{w}$. The surrogate is:

$$\hat{f}(\mathbf{w}; \mathcal{D}) = \mathbf{y}^\top (K + \lambda I)^{-1} k(\mathbf{w})$$

where $K_{ij} = \kappa(\phi(\tau_i), \phi(\tau_j))$, $\mathbf{y}_j = \mathbf{w}_j^\top J(\pi)$, and λ is a regularisation constant. The attribution of trajectory τ_j for preference $\mathbf{w}$ is the leave-one-out change in $\hat{f}$.

Learning or Inference Procedure

Each training step:

1. Sample a batch of preference weights $\{\mathbf{w}_j\}_{j=1}^B$ from the prior distribution over Δ^{K-1}.
2. Collect trajectories by rolling out each π_k under its currently assigned preference weights.
3. Compute attribution scores; upsample high-attribution trajectories.
4. Route trajectories to basis policies and apply PPO updates.

At test time, given a user preference $\mathbf{w}^*$, the system evaluates $J_k(\pi_k)$ for each basis policy and returns the best.

What the Guarantee Says

The regret bound $O(\sqrt{T \cdot c_{\max}})$ provides an instance-dependent characterisation: MORL is hard when objectives conflict strongly ($c_{\max}$ large) and easy when they are compatible ($c_{\max}$ small). The routed ensemble achieves this rate for any $K \geq c_{\max}/\epsilon$ for tolerance ϵ. This is the first formal characterisation of MORL hardness in terms of objective geometry.

Experimental Findings

- **Multi-objective MuJoCo locomotion:** Routed ensemble outperforms conditioned single-policy baselines by 8–15% in hypervolume metric across 4 multi-objective task suites.
- **Versus naïve ensemble:** Attribution-based curriculum sampling improves sample efficiency by ∼30% (same hypervolume with fewer environment interactions).
- **Conflict coefficient:** Empirical $c_{\max}$ correlates strongly with the observed performance gap between single-policy and ensemble methods ($r = 0.81$), validating the theoretical predictor.
- **Scalability:** Performance improvements hold for $K \in \{2, 4, 8\}$ basis policies; gains plateau at $K = 8$.

Ablations and Interpretation

Routing vs. random assignment: Random assignment of trajectories to basis policies reduces hypervolume by 7%, confirming that conflict-aware routing is the key driver.

Attribution vs. uniform sampling: Replacing kernel attribution with uniform trajectory sampling costs ∼20% in sample efficiency, isolating the contribution of the surrogate.

Conflict coefficient threshold: Task suites with $c_{\max} < 0.1$ show no significant gain from ensembles; the single-policy baseline already matches. This provides a practical criterion for when ensembles are needed.

Reference

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